□□□□

□□□□

MCR-BR-J

F1

AAEC CDCP CDCP

Abstract

End-to-end argument mining jointly identifies component spans, component types, and argumentative relations from raw text. Components are interdependent: paraphrasing a component during training may alter its relations to surrounding components, while changing its span or type during inference may invalidate existing relations. Increasing the number of textual variants or editing predictions independently does not explicitly address these dependencies. Building on the author's prior work on conclusion-specific modeling, multi-candidate reranking, and bounded repair, this thesis develops two complementary methods for preserving relational supervision and restoring prediction dependencies.

The first method is relation-aware augmentation learning for conclusion components. A conclusion-centered neighborhood defines the editable text, while internal relations, boundary-crossing relations, and annotated no-relation states define the affected supervision. Only designated component text may be rewritten. A deterministic reconstruction procedure transfers annotations to the augmented document, and detected semantic drift leads to rejection of the entire variant. In addition to the inherited adapter, cue-aware prompting, and sentence-level supervision, paired component and relation objectives encourage the shared generation representation to remain consistent across accepted variants. Source-document normalization controls the repeated contribution of unchanged content.

The second method, MCR-BR-J, performs dependency-closed argument-graph repair after multi-candidate reranking. A fixed candidate universe supports component-set replacements, including insertion, deletion, span or type changes, splitting, and merging. Every replacement is evaluated together with the reconstruction of all affected incoming and outgoing relations and previously absent links. The complete transaction is committed or rolled back atomically. An object-correctness scorer trained on out-of-fold predictions supplies a fixed count-based utility with a cost-sensitive risk interpretation. Strict acceptance and explicit budgets bound the search. Conditional analyses establish annotation transport, local utility differences, and finite termination, without claiming guaranteed improvement in true F1.

The evaluation protocol separates the contributions of training-time learning and inference-time repair on AAEC and CDCP, retaining CDCP multi-premise relations as native hyperedges. It includes related augmentation methods, matched-data loss ablations, sequential and joint repair under identical candidates and scores, global structured decoding, candidate recovery, harmful edits, and computational cost. Published prior results and lightweight diagnostics establish the research basis and its limits; empirical claims for the new methods require their own reproducible runs and strict evaluation.

Keywords: End-to-End Argument Mining; Relation-Aware Augmentation; Conclusion Components; Dependency Closure; Graph Repair; Cost-Sensitive Scoring

A

B

C

D

E

1.1 研究背景与问题提出

[1]

AAEC MajorClaim Claim Premise CDCP Fact Testimony Value Policy Reference [2] [3]

Claim

Claim

1.2 国内外研究现状

1.2.1

Stab Gurevych AAEC [2] Eger [4] Morio [5]

Caputo 2026 [6]

F1

1.2.2

Kawarada [7] Cabessa PE AbstRCT CDCP

[8] ACC

Das AASP

[9] 2025 arXiv AASP “ ”

2026 ARGUS [10] AAAI

Li LLM

[11] “ ”

1.2.3

Sun DENIM [12]

2025 Das

[13] Sun FOL-AM

[14] F1

FOL-AM

1.2.4

[15] CoDa

[16] SegMix [17] “ ”

Bao QOS[35] QOS

ConsistRE [38] CovEReD

[37] ODDA [39]

“ ”

Wang 2026 LLM [36]

“ ”

—

Rocha [40]

Sun

[18]

“ ”

QOS

1.2.5

token [19] JSON

AASP

Self-Refine [20] Huang

[21] [22]

Tran LLM-refine Claim Evidence Warrant

LLM [23] “

”

Xiao Li 2026 ARGUS LLM

[24] Caputo AAAI

Re-Cent [43]

Pirozelli 2026 [44]

“ ” “ ”

F1 Waegeman

F1 [41] Lipton F1 F1

[42] [45]

F1

ICIC 2026 MCR-BR[34]

F1 -only

1.2.6

Morio MT-AM [25]
“ ”

Savigny Yun 2026 AMELIA 19
[26]

Feger 2025 17 [27]
MCR-BR Cue-aware

ORCHID AAEC
[28] IAM [29] AAEC CDCP

“ _ ” “ ”
ICIC 2026 [34]

1.3 研究问题与主要研究工作

—

1.3.1

RQ1 NONE
“ ”

RQ2 —

AAEC CDCP

1.3.2

1

NONE “

”

2

“ ”

1-1

研究问题	关键机制	必要对照	结论边界
RQ1		QOS	F1
RQ2		ILP	F1
	× 2×2		

1.3.3 ICIC 2026

Improving End-to-End Argument Mining with LLMs via Multi-Candidate Reranking and Bounded Repair Chen Tang Min Peng Gang Tian ICIC 2026 Posters Volume I[34]

1-2

研究层次	**ICIC** 已公开机制	本论文新增内容	结果归属
			ICIC
			ICIC

ICIC “ ”

MCR-BR-C MCR-BR-J

1.4 □□□□□□

—　　—

2.1 论辩组件、二元关系与原生超边

$D=(x_1,\ldots,x_n)$ $c=(s,e,t)$ $1\le s\le e\le n$ 1

t token token

$r=(c_i,c_j,\ell)$ $G=(C,R)$ “

”

CDCP $h=(S,c_j,\ell)$ S c_j

1 R

[3][46]

2-1

语料	组件类型	关系对象	额外边界
AAEC	MajorClaim Claim Premise	Support Attack	Claim MajorClaim stance
CDCP	Fact Testimony Value Policy Reference	Reason Evidence	Claim/Premise

[2][3][46] CDCP

2.2 合法结构与语义身份

$\mathcal{A}$ $\mathcal{G}_{\mathcal{A}}$

JSON

AAEC

NONE stance CDCP Reason Evidence NONE

token “Libraries / should / stay / open / longer / . / Students / need / study / space / .”

```
{
  "components": [
    {"id": 1, "type": "Claim", "span": [1, 5]},
    {"id": 2, "type": "Premise", "span": [7, 10]}
  ],
  "relations": [
    {"sources": [2], "target": 1, "type": "Support"}
  ]
}
```

sources CDCP

ID type span

2.3 生成建模与参数高效适配

$Y=(y_1,\ldots,y_L)$

$$P_\theta(Y\mid D)=\prod_{t=1}^{L}P_\theta(y_t\mid D,y_{<t}). \quad 2\text{-}1$$

Transformer [30] Qwen2.5-7B-Instruct revision [31]

LoRA [32]

$$W=W_0+\frac{\alpha}{r}BA,A\in\mathbb{R}^{r\times d_{in}},B\in\mathbb{R}^{d_{out}\times r}. \quad 2\text{-}2$$

r α LoRA [33]

2.4 候选空间、评分与搜索

$$G_{rank}=arg\,max_{G\in\mathcal{S}}E_{rank}(G),G_0=L_{\mathcal{A}}(G_{rank}). \quad 2\text{-}3$$

$L_{\mathcal{A}}$

Candidate Oracle

U $\mathcal{V}$ P “ ”

2.5 严格评价与可比性

$$TP \quad FP \quad FN$$

$$P=\frac{TP}{TP+FP}, R=\frac{TP}{TP+FN}, F_1=\frac{2TP}{2TP+FP+FN}. \quad 2\text{-}4$$

micro-F1 micro-F1

CDCP F1

Relation-F1 Link-F1

FP

FN ID

micro-F1 F1 F1 F1 “F1 ”

“±”

2.6 两个阶段的闭环边界

“ — — — — ” “ — — —

”

→ →

→ → MCR

→ →

→ →

2.7 本章小结

3.1 研究问题与前期基础

C_{hit} R_{reach} C_{hit}

$|R|>0$ $Recall_{rel}\le|R_{reach}|/|R|$

ICIC 2026 [34] –

QOS [35] ConsistRE

[38] Wang [36] “ ” “ ”

“ ” AAEC MajorClaim Claim Premise CDCP Policy Value Fact Testimony Reference Policy/Value AAEC Claim AAEC stance MajorClaim

3.2 结论专项生成表示

3.2.1

Qwen2.5-7B-Instruct LoRA [32] $W_0\in\mathbb{R}^{d_{out}\times d_{in}}$

$$h'_t=W_0h_t+\frac{\alpha}{r}B_gA_gh_t+g_t\frac{\alpha}{r}B_cA_ch_t,A_\cdot\in\mathbb{R}^{r\times d_{in}},B_\cdot\in\mathbb{R}^{d_{out}\times r}. \quad 3\text{-}1$$

$g_t\in\{0,1\}$ id type span AAEC Claim/MajorClaim CDCP Policy/Value $g_t=0$

“ ” $r=16$

$\alpha=32$ rank-32 rank-16 rank-32

3.2.2

[33] therefore thus in conclusion consequently hence to conclude $P_i \in \mathbb{R}^d$

[40]

token EOS

$$L_{gen}(x,G) = -\frac{1}{T_y}\sum_{t=1}^{T_y}\log p_\theta(y_t / x, y_{<t}). \quad 3\text{-}2$$

ICIC

sigmoid $\hat{z}_i$ $z_i=1$

$$L_{sent} = -\frac{1}{N_s}\sum_{i=1}^{N_s}\left[z_i \log \hat{z}_i + (1-z_i)\log(1-\hat{z}_i)\right], L_{ICIC} = L_{gen} + \lambda_{sent} L_{sent}. \quad 3\text{-}3$$

$\lambda_{sent}=0.5$

3.3 □□□□□□□□□□□□□□

3.3.1

$G=(C,R)$ A

$\Omega(C)$ $\omega=(S,v)$ S v AAEC $|S|=1$

y_ω NONE

CDCP $(\{c_{i_1},\ldots,c_{i_k}\},c_j,\ell)$ $\ell \in \{Reason, Evidence\}$

NONE

NONE

$V(\omega)=S\cup\{v\}$

$$\mathcal{D}_A=\{\omega\in\Omega(C):V(\omega)\cap A\neq\varnothing\}. \quad 3\text{-}4$$

$\{(u,v):u\in A \vee v\in A,(u,v)\in\Omega(C)\}$ A A

NONE “ ”

3.3.2

“ NONE”

ID

$$\varphi:C\longrightarrow C',(s_c,e_c,t_c)\longmapsto(s'_c,e'_c,t_c). \quad 3\text{-}5$$

ID φ

token φ $\varphi(G)$

c_1 c_2 c_2 c_3 $c_2\to c_1$ $c_2\to c_3$

NONE c_2 c_1 c_3

3.3.3

NONE

φ token

d $N_{d,\tau}$ τ a_d

$N'_\tau=N_\tau+\sum_d a_d N_{d,\tau}$

[39]

3.4 □□□□□□□□□□□□□

3.4.1

h_i^x

offset mapping $B_x(c)$

h_{end}^x

$$u_c^x = MeanPool_{i \in B_x(c)} h_i^x, p_C^x(c) = softmax f_C([u_c^x; h_{end}^x]). \quad 3\text{-}6$$

token h_{end}^x

token

$$p_R^x(S, v) = softmax f_R([Pool(\{u_c^x : c \in S\}); u_v^x; h_{end}^x; \eta(S, v)]), \quad 3\text{-}7$$

f_C f_R $Pool$ η

token

NONE “ ” “ ”

3.4.2

$$A^+ = A \cup \bigcup_{\substack{\omega \in \mathcal{D}_A \\ y_\omega \neq \text{NONE}}} V(\omega). \quad 3\text{-}8$$

NONE x'

$$L_C^{align}(x, x') = \frac{1}{|A^+|} \sum_{c \in A^+} JS(p_C^x(c), p_C^{x'}(\varphi(c))), \quad 3\text{-}9$$

$JS(p, q) = \frac{1}{2} KL(p \| \overline{p}) + \frac{1}{2} KL(q \| \overline{p})$ $\overline{p} = (p + q)/2$

D_+ NONE D_0 $\mathcal{B} = \{b \in \{+, 0\} : |D_b| > 0\}$

$$L_R^{align}(x, x') = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \frac{1}{|D_b|} \sum_{\omega \in D_b} JS(p_R^x(\omega), p_R^{x'}(\varphi(\omega))). \quad 3\text{-}10$$

NONE $\mathcal{D}_A = \varnothing$

A^+

JS

NONE $S(x,G)$

3.4.3

d m_d

$$L_{heads,d}^{sup}=\frac{S(x_d,G_d)+\sum_{j=1}^{m_d}S(x'_{dj},\varphi_j(G_d))}{1+m_d}. \quad 3\text{-}11$$

m_d $\bar{L}_{C,d}^{align}$ $\bar{L}_{R,d}^{align}$

$$L_d= \quad L_{ICIC}(x_d,G_d)+\rho\frac{1}{m_d}\sum_{j=1}^{m_d}L_{ICIC}(x'_{dj},\varphi_j(G_d)) \quad 3\text{-}12$$

¿ ¿

$m_d=0$ ρ

LoRA

3.5 算法过程与性质

3-1

算法 **3-1** 关系感知增强学习

1. $\Omega(C)$
2. A $\mathcal{D}_A$
3. φ
4.
5.
6.
7.

φ φ G $\varphi(G)$

y p $\delta = p_y - max_{k \neq y} p_k > 0$

q $\| p - q \|_\infty < \delta / 2$ $arg\,max\, q = y$ $k \neq y$ $q_y - q_k > p_y - p_k - \delta \geq 0$

JS

3.6 独立实验协议

ICIC

QOS

NONE JS

/

micro-F1 micro-F1 CDCP

CovEReD

[37] JS

3.7 本章小结

ICIC

MCR-BR —

MCR-BR-J J

4.1 研究问题与前期方法的继承

ICIC

“ ” BR-O

BR-C [34]

BR-C

BR-O “

”

[46] Re-Cent

“ ” [43]

[44] MCR

x G_0 G

F1

4.2 多候选完整图与初始选择

4.2.1

x K

$$\mathcal{C}=\{C^{(1)},\ldots,C^{(K)}\}. \quad 4\text{-}1$$

$C^{(k)}$ $G^{(k)}=(C^{(k)},R^{(k)})$

4.2.2

MCR

$$E_{rank}(G)=0.4E_{dens}(G)+0.4E_{coh}(G)+0.2E_{val}(G). \quad 4\text{-}2$$

$\mathcal{Q}(G)$ $p_r\in\{0,1\}$ $d_{coh}^{\text{in}}(j)$ j

$$E_{dens}(G)=\frac{1}{|\mathcal{Q}(G)|}\sum_{j\in\mathcal{Q}(G)}min\left\{\frac{d_{coh}^{\text{in}}(j)}{\tau_j},1\right\},E_{coh}(G)=\frac{1}{|R|}\sum_{r\in R}p_r. \quad 4\text{-}3$$

AAEC Claim CDCP Policy/Value MajorClaim

$\tau_j=2$ $\tau_j=1$

CDCP

E_{val} 1 0

G_0

ID ID

p_r

4.3 □□□□□□□□□□□□□

4.3.1

U K

U

$u\in U$ NULL $\mathcal{V}=\{(u,t):u\in U,t\in\mathcal{T}(u)\}$

$C\subseteq\mathcal{V}$

“ ”

token

token

offset mapping

4.3.2

$\mathcal{V}$ P P

NONE

“ ” “ ”

4.4 冲突闭合的集合替换与关系事务

4.4.1

C B D_0 $B \cap C$ B D_0

$$D = D_0 \cup \{ c \in C : \exists\, b \in B, Conflict(c, b) \}, C' = (C \setminus D) \cup B. \quad 4\text{-}4$$

B D

$0 \to 1$ $1 \to 0$ $1 \to 1$ $1 \to 2$ $2 \to 1$

$|D| + |B| \le 3$

4.4.2

$A = D \cup B$ P_A P A

NONE

S_R

$$P_{tx} = P_A \cup S_R. \quad 4\text{-}5$$

RelationOnly $D = B = \varnothing$ S_R S_R

C' $G' = (C', R')$ C'

4.5 完整对象评分及训练外候选学习

4.5.1

$$q_C(c \mid x), q_R(r \mid x). \quad 4\text{-}6$$

NULL/

MLP MLP

q

4.5.2 OOF

out-of-fold OOF

OOF

NONE

OOF OOF

4.6 □□□□□□□□□□□□

4.6.1

$X \in \{C, R\}$ $0 < \kappa_X < 2$

$$\Phi_X(G) = 2\sum_{o \in X(G)} q_X(o/x) - \kappa_X |X(G)|, \Phi(G) = w_C \Phi_C(G) + w_R \Phi_R(G), \quad 4\text{-}7$$

$w_C, w_R > 0$ $w_C = w_R = 1$ κ_X F1

Y_X $q_X(o/x) = P(o \in Y_X / x)$ $FP_X = |X(G)| - TP_X$ $FN_X = |Y_X| - TP_X$

$$\Phi_X(G) = (2 - \kappa_X)\mathbb{E}[|Y_X|/x] - \mathbb{E}[\kappa_X FP_X + (2 - \kappa_X) FN_X / x]. \quad 4\text{-}8$$

F1 F1

[41] F1 F1 [42]

4.6.2

$$\Delta\Phi_C \geq 0, \Delta\Phi_R \geq 0, \Delta\Phi > \delta > 0. \quad 4\text{-}9$$

F1 δ

$d_C = |D| + |B|$ $d_R = |R \triangle R'|$

“ ” G_0

4.7 □□□□□□□□□□□□□

4.7.1

“ ILP ” C' $z_c = \mathbb{I}[c \in C']$ $y_r \in \{0,1\}$ $r \in P$

$$max_y \ w_C \Phi_C(C') + w_R \sum_{r \in P} (2 q_R(r/x) - \kappa_R) y_r. \quad 4\text{-}10$$

$$\begin{aligned} &\sum_{t \in \mathcal{T}(u)} z_{(u,t)} \leq 1, && ¿\, z_c + z_d \leq 1 \quad Conflict(c,d); \\ &y_r \leq z_c \ \forall c \in Ends(r), && ¿\, \sum_{r \in \Gamma_s} y_r \leq 1 \ \forall s; \\ &y_r = \mathbb{I}[r \in R] \ \forall r \notin P_{tx}, && ¿\, G' \in \mathcal{G}_{\mathcal{A}}. \end{aligned} \quad 4\text{-}11$$

Γ_s NONE $\mathcal{G}_{\mathcal{A}}$

$0 \leq o_c \leq M - 1$ $M = |\mathcal{V}|$

$$o_v \geq o_u + 1 - M(1 - y_{(u,v,\ell)}). \quad 4\text{-}12$$

CDCP

C' $y_h = 0$

ILP $y_r^0 = \mathbb{I}[r \in R]$

$$d_R = \sum_{r: y_r^0 = 1} (1 - y_r) + \sum_{r: y_r^0 = 0} y_r. \quad 4\text{-}13$$

d_C, d_R $\Delta\Phi_C, \Delta\Phi_R \geq 0$

$\Delta\Phi > \delta$

4.7.2

32

3

4-1

```
     x K
1.    MCR              G0 G ← G0
2.       U P
3.                  no-op
4.         (D0, B, SR)
   a.                 C'
   b.           NONE
   c.            ILP
   d.
5.

6.                  3
7.
```

no-op ILP “ILP ”

incumbent

4.8 □□□□□□□□□

4.8.1

4-1

$X \in \{C, R\}$ $a_X(o) = 2q_X(o/x) - \kappa_X$

$$\Delta\Phi_X = \sum_{o \in X(G') \setminus X(G)} a_X(o) - \sum_{o \in X(G) \setminus X(G')} a_X(o). \quad 4\text{-}14$$

4.8.2

4-2

a, b c $\kappa_C = 0.4$ $q_C(a) = q_C(b) = 0.3$ $q_C(c) = 0.6$

$0.2, 0.2, 0.8$ $0.3\{a,b\} + 0.6\{c\} + 0.1\varnothing$ $\{a,b\}$ 0.4

c $\{a,b\}\rightarrow\{c\}$ 0.4 $w_C=1$ $0<\delta<0.4$

F1

4.8.3 F1

4-3 Φ

micro-F1 T P_0 Q $F=2T/(P_0+Q)$

$$F'-F=\frac{2\Delta T-F\Delta P_0}{P_0+Q+\Delta P_0}. \quad 4\text{-}15$$

ΔT F κ

F1 F1

4.9 □□□□□□□□□

AAEC MajorClaim Claim Premise Claim stance Support/Attack MajorClaim

CDCP

$$h=(\{c_{i_1},\ldots,c_{i_k}\},c_j,\ell),\ell\in\{Reason,Evidence\}. \quad 4\text{-}16$$

[46]

3

ILP

ILP z_c h c $y_h+z_c\leq 1$ ILP C'

MCR-only BR-O BR-C RelationOnly

RelationOnly BR-C

ILP

BR-O BR-C

4.10 □□□□

MCR

5.1 验证目标与证据分层

ICIC CPU

5-1

证据类别	当前材料	可支持范围	不能支持的结论
ICIC		MCR-BR	
			run ID
CPU			LLM

5.2 数据、划分与运行身份

5.2.1

AAEC Support/Attack stance CDCP Reason/ Evidence ID

ID

5.2.2

checkpoint

—

token token CPU

ILP ILP gap “ ” “

”

5.3 第一项研究：增强与学习的可识别比较

5.3.1

T0 ICIC

QOS QOS [35]

“

” QOS

NONE

5.3.2

“ ”“ / ”“ ”

5-2

对照	固定条件	变化因素	验证问题
T0			
QOS			
		NONE	

— P/R/F1

“therefore”

5.3.3

/ /NONE

ID

5.4 第二项研究：候选、评分与闭合修正

5.4.1

Oracle

“Oracle”

Brier

TP/FP/FN

5.4.2

MCR-only BR-O ICIC BR-C BR-J O C J

5-3

对照	对齐要求	主要问题	必须报告的代价
MCR-only BR-O BR-C BR-J			
			Recall
ILP			gap
	token		

ILP

ILP

F1

5.4.3

“ ”

TP/FP/FN

5.5 组合设计、统计与决策标准

T0/T1 I0/I1 2×2 T0 ICIC T1 I0 MCR-only I1 BR-J I0/I1 “ ”

3 5

micro-F1

5-4

研究判断	支持所需证据	不满足时的处理
	–	
		ICIC

5.6 已有结果及轻量诊断的解释

5.6.1 ICIC

ICIC Full AAEC F1 83.71 F1 64.83 CDCP 76.45 38.17[34]

CDCP

Oracle E ICIC
Single-Pass F1 81.67 77.82 run ID
BR-C BR-J

5.6.2 CPU

AAEC 322 191 65 66 CDCP
00411 580 348 116 116 AAEC
CDCP

typed component F1 AAEC 25.42 CDCP 4.89
CDCP LLM ILP

LLM

5.7 跨语料补充协议

AAEC CDCP AAEC MajorClaim Claim Claim
Premise CDCP Policy Value Claim Fact Testimony Reference Premise

CDCP “ ”
Relation-F1
CDCP
CDCP E

5.8 本章小结

Full

6.1 研究总结

ICIC

NONE

ICIC

F1

AAEC CDCP

6.2 局限性

NONE

ILP

CPU LLM

6.3 后续工作

LLM

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A

“ ”

A-1

编号	对应事项	最低材料
R01		/
R02		ID
R03		token
R04		
R05	E	81.67 77.82 83.71 run ID
R06		ID
R07		ICIC
R08		
R09		“±”
R10		
R11		
R12	Oracle	
R13		/
R14		
N01	CDCP	
N02		
N03		OOF /
N04		no-op
N05		/ 2×2 —

B

B.1 数据与模型身份

run ID ID / token revision

ID

B.2 评分与接受轨迹

MCR

JSON

“ ” TP/FP/FN

B.3 no-op、失败及预算

/

“ ”

B.4 自动测试的最低覆盖

ID NONE

ILP

C

C.1 前期共同成果

Chen Tang Min Peng Gang Tian ICIC 2026 [34]
1-2 E

C.2 辅助工具与责任边界

“ ”

D

D.1 既有工程身份

argument_mining_0914 ICIC 0915
75 Qwen2.5-7B

research_0920 CPU AAEC/CDCP BR-J
“ ”

D-1

机制	已有基础	本版仍需的实现与验证
	/	/NONE
O/C		
	CPU	LLM /
ILP		
CDCP		expand

D.2 参数来源与复现范围

Qwen2.5-7B-Instruct rank 16 alpha 32 0.5 K=3 MCR 0.4/0.4/0.2
epoch batch ICIC

E

0915-v1 E-1 ICIC “ ” 0920-v1 “±” A

E.1 AAEC □□□□□□□□□

E-1 AAEC

□□	MajorClaim-F1	Claim-F1	Comp-F1
FLAN-T5 (XXL)	71.23	76.45	80.15
LLM Single-Pass	73.45	78.21	81.67
Ours (Full)	**77.34**	**81.77**	**83.71**

ICIC Single-Pass 81.67 E-2 77.82 run ID

E.2 ICIC □□ AAEC □□□□□□

E-2 ICIC AAEC

□□	Comp-F1	Support-F1	Attack-F1	Relation-F1
FLAN T5-XXL	80.15	62.34	48.72	61.19
GPT-4-turbo 20-shot	55.51	29.86	18.45	28.38
RoBERTa-large MTL	76.43±0.55	54.21±0.52	41.83±0.67	52.15±0.54
LLM Single-Pass	77.82±0.51	58.43±0.45	44.21±0.62	56.87±0.48
LLM w/ Constrained Decoding	78.54±0.46	59.12±0.42	45.33±0.58	57.65±0.45
LLM w/ DAG Post-process	78.91±0.49	59.45±0.48	45.87±0.61	58.12±0.47
LLM Multi-Sample (LogProb)	79.45±0.38	60.12±0.41	46.58±0.55	58.94±0.42
LLM w/ Self-Refine	79.82±0.45	60.75±0.43	47.12±0.56	59.63±0.45
Ours ($K=1$, w/o correction)	81.24±0.44	61.87±0.39	48.34±0.58	60.73±0.41
Ours (Verifier-only Rerank, $K=3$)	82.18±0.40	63.15±0.36	49.82±0.52	62.21±0.38
Ours (Full, $K=3$)	**83.71±0.42**	**65.92±0.38**	**51.47±0.54**	**64.83±0.38**

ICIC Table 1 AAEC [34]

E.3 ICIC 原版 CDCP 域内结果记录

E-3 ICIC CDCP

方法	Comp-F1	Reason-F1	Evidence-F1	Relation-F1
FLAN T5-XXL	72.68	35.24	28.41	33.96
RoBERTa-large MTL	68.52±0.58	31.45±0.61	25.72±0.73	30.23±0.64
LLM Single-Pass	71.35±0.48	33.82±0.52	27.15±0.68	32.64±0.55
LLM w/ DAG Post-process	72.14±0.50	34.65±0.49	28.02±0.63	33.51±0.52
LLM w/ Self-Refine	72.95±0.47	35.58±0.48	28.94±0.60	34.42±0.50
Ours ($K=1$, w/o correction)	74.56±0.45	36.94±0.46	30.27±0.58	35.78±0.48
Ours (Verifier-only Rerank, $K=3$)	75.12±0.41	37.85±0.44	31.42±0.55	36.64±0.45
Ours (Full, $K=3$)	**76.45±0.43**	**39.12±0.45**	**32.86±0.57**	**38.17±0.46**

ICIC Table 1 CDCP [34]

E.4 ICIC 原版模块消融记录

E-4 ICIC

配置	Comp-F1	Relation-F1	ΔComp	ΔRel
Full Method	83.71±0.42	64.83±0.38	—	—
$K=1$, w/o correction	81.24±0.44	60.73±0.41	-2.47	-4.10
w/o Claim-specific Adapter	81.96±0.48	63.08±0.44	-1.75	-1.75
w/o Cue-aware Prompting	82.34±0.45	62.87±0.40	-1.37	-1.96
w/o Multi-Objective	82.89±0.43	63.54±0.39	-0.82	-1.29
w/o Validity Signal E_{val}	81.56±0.46	61.92±0.43	-2.15	-2.91

ICIC Table 2[34]

E.5 ICIC 原版验证器替换记录

E-5 ICIC

验证器	Comp-F1	Relation-F1	Top-1 一致率	修正触发率
Qwen2.5-7B	83.71	64.83	100.0%	42.5%
Llama-3-8B	83.71	64.55	94.2%	43.8%

ICIC Table 3[34]

E.6 ICIC 原版 Oracle 记录

E-6 ICIC Oracle

条件	Comp-F1	Relation-F1	ΔRel
	83.71±0.42	64.83±0.38	—
Candidate Oracle $K=3$	85.12±0.36	67.24±0.34	+2.41
Condition Oracle	—	78.42±0.35	+13.59
Output Oracle	100.00	83.35±0.32	+18.52

ICIC Table 5[34]

E.7 ICIC 原版修正行为记录

E-7 ICIC

指标	结果
	42.5%
	1.8
	2.4
LocalRerun	45.2%
RemoveEdge	24.8%
Replace	19.5%
Promote/Demote	10.5%
ID	18.5% → 2.1%
JSON	6.2% → 0.0%
	22.4% → 5.8%
E_{val}	75.3% → 97.9%

ICIC Table 6[34] Replace Promote/Demote BR-O BR-C BR-J

E.8 前期粗粒度组件映射

E-8

数据集原标签	粗粒度标签	映射依据
AAEC MajorClaim	Claim	
AAEC Claim	Claim	
AAEC Premise	Premise	
CDCP Policy	Claim	
CDCP Value	Claim	
CDCP Fact	Premise	
CDCP Testimony	Premise	
CDCP Reference	Premise	

AAEC/CDCP

E.9 ICIC 原版 AAEC 到 CDCP 迁移记录

E-9 ICIC AAEC CDCP

方法	**Coarse Component-F1**	**Unlabeled Link-F1**
FLAN T5-XXL zero-shot	52.34±0.65	31.25±0.72
LLM Single-Pass zero-shot	54.87±0.58	33.42±0.68
Ours ($K=1$, w/o correction, zero-shot)	57.23±0.52	38.76±0.61
Ours (Full, zero-shot)	**60.78±0.48**	**43.21±0.55**

ICIC Table 4[34] Link-F1 F1